

A Study Of the Concepts, Capabilities, and Limits of Quantum Computing

Conference Tutorial

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Quantum computing is a model of computation that uses principles from quantum mechanics to process information [1]. Classical computation begins with the bit, which stores a value of either 0 or 1. Quantum computers instead use *qubits*, whose states can exhibit superposition, entanglement, and interference. These properties make quantum computers useful for certain specialized problems, including integer factorization and structured search, where quantum algorithms can offer advantages over known classical approaches [2].

This hands-on tutorial offers a practical introduction to quantum computing for undergraduate computing students, technically curious faculty, and IT professionals. No prior background in quantum mechanics is required although familiarity with basic computer science, linear algebra, and probability will be helpful.

Participants will use a Jupyter Notebook [3] built around IBM's Qiskit framework [4, 5]. The notebook uses Qiskit to construct small quantum circuits, run them on simulators, and interpret the resulting measurement counts. These examples allow participants to see how quantum programs are represented, executed, and analyzed without requiring access to physical quantum hardware.

The session begins by comparing the classical bit with the qubit, the basic unit of quantum information. Participants then explore measurement, superposition, probability amplitudes, and entanglement through small circuit examples. The presentation avoids unnecessary physics and instead frames quantum behavior in terms of information representation and computation.

The tutorial next introduces quantum gates and circuits as counterparts

to classical logic gates and circuits. Examples using the Hadamard gate and the controlled-NOT gate show how quantum states are transformed and how simple circuits can produce superposition and entanglement. The session then briefly connects these ideas to landmark algorithms, including Grover’s search algorithm [6] and Shor’s factoring algorithm [7].

The final portion addresses the practical limits of current quantum computing. Participants consider what quantum computers may do well, what they do not yet do well, and why noise, decoherence, limited qubit counts, and error correction remain central challenges. The discussion also connects quantum computing to likely application areas, including cryptography, optimization, quantum simulation, and quantum-safe security [8].

By the end of the tutorial, participants should be able to distinguish a classical bit from a qubit, describe superposition and entanglement in plain language, explain the role of quantum gates and circuits, identify several important quantum algorithms, and separate realistic near-term uses of quantum computing from common misconceptions.

Availability

The Jupyter Notebook and supporting files for this tutorial are hosted on GitHub at https://github.com/adamwadelewis/quantum_tutorial.

Biography

Dr. Lewis joined Athens State University’s Division of Mathematical, Computer, and Applied Sciences in the Fall of 2012 after an extensive career in the computing industry. His research interests include computational linguistics, high-performance computing, and the computer architectures and systems that support those fields.

References

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